

Elijah Parsons

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EDUCATION

Purdue University West Lafayette, IN | May 2028
Bachelor of Science, Mechanical Engineering with Manufacturing Concentration GPA (3.96/4.00)
Lilly Endowment Community Scholarship Finalist, Dean's List and Semester Honors
Relevant Skills: 3D Modeling – Onshape, Inventor, NX; Programming – python, Java, C, Excel

PROFESSIONAL EXPERIENCE

Metal Technologies Incorporated | Engineering Intern | Bloomfield, IN May 2025 – August 2025

- Analyzed 1,000+ tool-failure records over 3+ years, sorting failures by tool and CNC machine to identify recurring patterns; identified 4–5 high-cost, frequently failing tools and presented findings to management.
- Repaired and maintained 12 aluminum CNC machines used to machine engine blocks; diagnosed pneumatic part-detection issues, verified airflow, and calibrated pressure thresholds to prevent bad parts from passing.
- Developed Excel/VBA automation to digitize tool setup tracking, automatically archive checklists, and count 40+ daily setups, reducing manual work and improving setup tracking.
- Collaborated with technicians to create 20+ standardized tool setup instructions and checklists, improving setup consistency and reducing failures caused by improper setup.

PERSONAL PROJECTS

3D-Modeled 5×5 Rubik's Cube | Onshape, 3D Printing

- Reverse-engineered and modeled 8 custom components in Onshape, iterating geometry, clearances, and revolving cuts to resolve interference and enable smooth layer rotation.
- 3D-printed and tested components on a Prusa printer, optimizing print orientation and tolerances to produce a fully functional 5×5 cube.

Algorithmic Stock Trading Bot | Python, GitHub, NumPy, Backtrader, yfinance, excel

- Developed and backtested a short-term trading strategy by coding stock indicators across 500+ S&P 500 stocks with 5–6 years of historical data, achieving a 16% annualized return with 1% average gain per trade.
- Automated nightly stock screening and trade signals through GitHub, which sends buy/sell trades every day.

Bass Guitar Suitcase Amp | Onshape, Woodworking, 3D Printing, Wiring

- Designed a custom suitcase amplifier enclosure in Onshape, optimizing internal volume for a subwoofer, amp, and power distribution.
- Calculated wattage and impedance requirements to safely wire and integrate the amplifier with a car subwoofer.
- Engineered 3D-printed magnetic brackets to secure wireless receivers and a multi-effects pedal with quick-release manual detachment.

EXTRACURRICULARS

Pi Tau Sigma ME Honors Society | Fundraising Chair Aug 2025 – Present

- Aided Treasurer by finding opportunities to fundraise and provided details how to apply and work them.

FIRST Robotics Competition | Design Team Captain Aug 2022 – May 2024

- Led a 6-person design team in developing, prototyping, and testing a competition robot, resolving design flaws before final implementation.
- Created full-robot CAD models in Onshape of the design of the drivetrain, intake, arm, shooter, and climber.
- Assembled the frame and mechanical systems, and troubleshooted issues throughout testing and competition.

Worship Band – Bethany Christian Church | Bass Player Mar 2023 – Present

- Perform bass guitar for weekly church services and two weeklong summer camps.